

Disparity Map based Synthetic IR Pattern Augmentation for Active Stereo Matching

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CCS Concepts

• Computing methodologies → Shape inference.

Keywords

Data augmentation, Active stereo matching, Disparity estimation

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1 Introduction

Deep learning based stereo matching requires a large amount of training data, but datasets for active stereo matching are scarce and it is difficult to collect data in real-world environments. In this work, we propose a novel method for synthesizing infrared (IR) projector patterns from readily available passive stereo images. We applied the proposed method to the SGM [Hirschmuller 2008] algorithm and deep learning models to verify its performance and analyze the feasibility of using the passive stereo matching models for active stereo matching. The experimental results show a reduction in error rate of up to 51.7% for the SGM method and up to 40.4% for the deep learning based disparity estimation methods.

2 Synthetic IR Pattern Augmentation

Stereo cameras equipped with built-in IR pattern projectors suffer from pattern occlusion due to parallax effects caused by the spatial displacement between the projector and cameras.

As depicted in Figure 1, the IR pattern projected from the projector is occluded by Object B, resulting in missing patterns on regions R_3 , R_4 , and R_5 . Additionally, the left camera cannot see region R_6

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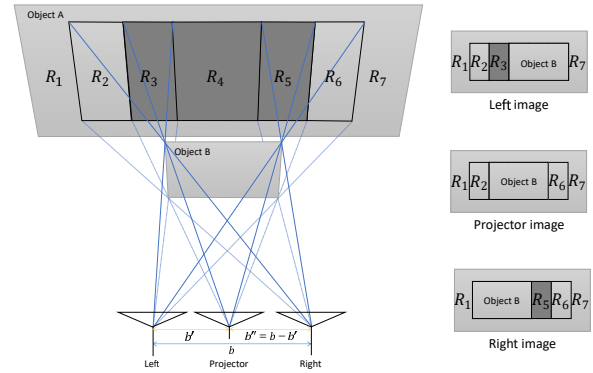


Figure 1: Occlusion areas caused by differences in the position of the projector

with the projected pattern while seeing region R_3 without the pattern, and the right camera cannot see region R_2 with the pattern but can see region R_5 without it.

Detecting such parallax-induced occlusion traditionally involves computationally intensive 3D space occlusion detection. To address this, we propose an efficient method to synthesize IR patterns using disparity maps without relying on explicit occlusion detection.

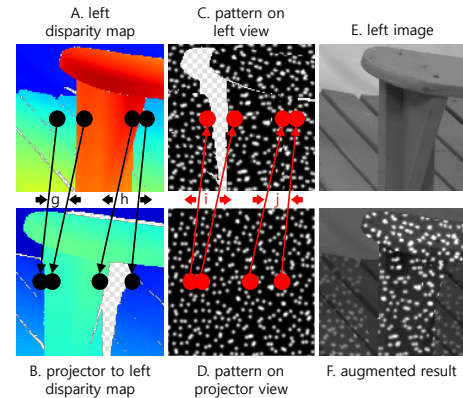


Figure 2: Synthetic IR pattern augmentation

Figure 2 illustrates our proposed image transformation process. Initially, the left disparity map in Figure 2.A is used to compute the

projector-to-left disparity map in Figure 2.B. This is achieved by multiplying the left disparity map by the baseline ratio b'/b from Figure 1 to obtain the left-to-projector disparity map, followed by a pixel-wise left shift based on the disparity values. Regions labeled g in Figure 2 correspond to areas without projected patterns, similar to R_3 in Figure 1, while regions labeled h represent areas with projected patterns invisible to the left camera, akin to R_6 . Subsequently, the projector pattern in Figure 2.D is used as a projector-to-left disparity map to generate the pattern from the left viewpoint in Figure 2.C. This involves shifting each brightness value in the pattern image to the right according to the disparity, with areas lacking disparity information being treated as occlusions. Finally, the pattern from the left viewpoint undergoes brightness scaling based on distance and is combined with the original left image in Figure 2.E to produce the final patterned image in Figure 2.F. The same process is applied to the right image to obtain the patterned right image.

3 Experiment

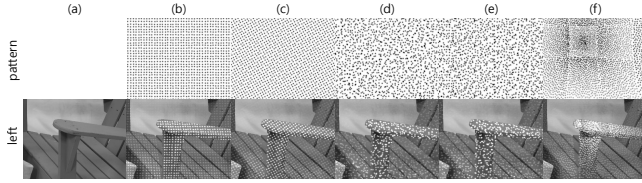


Figure 3: Patterns and augmented images

Figure 3 shows a crop of the synthesized images with the patterns used in the experiment. We implemented a total of six patterns for the IR projector: (a) grayscale to check the performance against unpatterned images, (b) and (c) the simplest possible dot grid pattern and a dot grid pattern rotated by 30° , (d) a pseudo-random pattern, (e) a pattern that repeats in the vertical direction similar to commercial products, and (f) a pattern that simulates the output of a module using a Vertical-Cavity Surface-Emitting Laser (VCSEL) and a Diffractive Optical Element (DOE).

In this work, the patterns were generated and synthesized so that about 10000 spots were drawn on the Middlebury 2014 [Scharstein et al. 2014] dataset, which is a passive stereo dataset with real data. The IR projector is set to be centered in front of the left and right cameras.

We use SGM [Hirschmuller 2008] algorithm, PSMNet [Chang and Chen 2018], RAFT-Stereo [Lipson et al. 2021], IGEV-Stereo [Xu et al. 2023], and IINet [Li et al. 2024] models trained with Sceneflow [Mayer et al. 2016] as disparity estimation method.

Figure 4 shows the results of estimating the disparity map with SGM and RAFT-Stereo by adding pattern (e) to the Recycle image. We can see that the error in the concavities is reduced after adding the pattern, indicating that pattern synthesis helps in estimating the disparity map.

The results in Table 1 show the impact of each pattern on disparity inference. The neural network (NN) methods reduce the error rate by 14.2% to 40.3% in (d) or (e) with pattern synthesis compared to (a) without pattern synthesis. In (b), NN methods except PSMNet have more errors, indicating the need for proper pattern

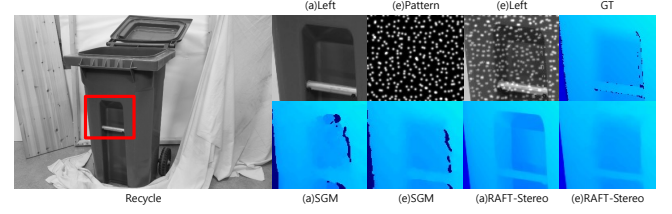


Figure 4: Recycle image and disparity map with pattern (a) and (e)

Table 1: Percentage of pixels with errors of 1 pixel or more

Pattern	(a)	(b)	(c)	(d)	(e)	(f)
SGM (%)	18.0 (0.0)	10.2 (-43.3)	8.8 (-51.4)	8.7 (-51.7)	8.7 (-51.9)	9.5 (-47.4)
PSMNet (%)	26.2 (0.0)	24.5 (-6.5)	19.4 (-25.8)	15.6 (-40.3)	15.8 (-39.7)	17.1 (-34.6)
RAFT-Stereo (%)	14.8 (0.0)	17.4 (18.1)	16.0 (8.3)	12.1 (-18.0)	11.9 (-18.9)	12.7 (-13.6)
IGEV-Stereo (%)	12.7 (0.0)	15.1 (18.9)	13.7 (7.7)	10.9 (-14.2)	11.1 (-12.7)	11.7 (-7.6)
IINet (%)	25.7 (0.0)	26.2 (1.8)	21.9 (-15.0)	18.7 (-27.1)	18.6 (-27.6)	21.0 (-18.3)
NN-Average (%)	19.9 (0.0)	20.8 (4.8)	17.7 (-10.6)	14.3 (-27.7)	14.4 (-27.6)	15.6 (-21.1)

design. SGM has a higher error rate compared to RAFT-Stereo and IGEV-Stereo in (a), but it has the lowest error rate in (b) to (f) with 43.3% to 51.9% lower error rate after applying patterns. The NN methods show lower generalization performance compared to SGM, suggesting that fine-tuning is needed.

Proposed method helps in disparity estimation, and confirmed that IR patterns can improve the performance of deep learning based passive stereo matching models without fine-tuning. Furthermore, improvements can be expected with proper pattern design and learning based on it.

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